

Integration — Lesson 13 Test: Real-World Accumulation Problems

10 questions · show your work in the box · calculators allowed · every answer needs UNITS

Name: _____ Date: _____ Score: _____ / 10

1. Water flows into a tank at the rate $r(t) = 6t$ liters per minute. Use an integral to find how much water enters during the first 3 minutes. 1 pt

2. A tutoring business earns money at the rate $r(t) = 8 + 2t$ dollars per hour. How much does it earn from $t = 0$ to $t = 5$ hours? 1 pt

3. A bacteria colony grows at the rate $r(t) = 3t^2$ bacteria per hour. How many NEW bacteria appear during the first 4 hours? 1 pt

4. A rain barrel already holds 15 liters. Water flows in at a steady 4 liters per minute for 6 minutes. How much water is in the barrel at the end? 1 pt

5. Sand pours onto a pile at the rate $r(t)$ kilograms per minute, where t is in minutes. In plain English, what does $\int_1^4 r(t) dt$ mean in this story, and what are its units? 1 pt

6. A tank starts empty and fills at the rate $r(t) = 8t$ liters per minute. At what time does it reach 100 liters? 1 pt

7. Water pours into a tub at 10 L/min while the drain lets water out at $1 + t$ L/min. (a) Write the net rate. (b) How much water does the tub gain in the first 6 minutes? 1 pt

8. A faucet runs at $r(t) = 2t$ L/min for the first 4 minutes, then holds a constant 8 L/min from $t = 4$ to $t = 7$. How much water flows out in total from $t = 0$ to $t = 7$? (Split the integral!) 1 pt

9. Shop A earns at a steady 30 dollars per day. Shop B earns at $8t$ dollars per day. Over the first 8 days, which shop earns more, and by how much? 1 pt

10. A dish starts with 100 bacteria, and the colony grows at $r(t) = 3t^2$ bacteria per hour. After how many hours does the population reach 316? 1 pt
